Pharmacology of
Drugs Used in Coagulation Disorders, Part 1
Anticoagulants

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Following this session, students should be able to:

1. List the pharmacokinetic properties of each class of drugs affecting blood coagulation and give specific examples.
2. Explain the structure-activity relationship of each of the classes and give specific examples.
3. Relate the mechanisms of action of each class of drugs to their effects on blood coagulation and give specific examples.
4. Explain the clinical use of each class of drugs and give specific examples.
5. Correlate the mechanisms of toxicities, drug-disease interactions and drug-drug interactions with the contraindications for use giving specific examples.

Preparation Materials (links are in the CPG and on the next slide)

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Required

- ScholarRx Bricks | Practice Questions | See next slide.

Highly relevant optional materials:

- Dr. Goldstein's Word handout | Video Lecture | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them.*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions (not graded): Simple Recall and Case Vignettes*

Links to resources

Scholar Rx Bricks:
Name of Discipline/ Organ System: Hematology > Hemostasis Disorders > Thrombotic Disorders
Title of Brick: Anticoagulant drugs
Link: https://exchange.scholarrx.com/brick/anticoagulant-drugs

Access Medicine Katzung & Vanderah's Basic & Clinical Pharmacology, 16e, 2024; Chapter 34: Agents Used in Disorders of Coagulation
https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3382&sectionid=281752540

LWW Health Libraries, Premium Basic Sciences: Lippincott's Illustrated Reviews, Pharmacology, 8e, 2023; Chapter 13: Anticoagulants and Antiplatelet Agents
https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253326640&bookid=3222

Access Medicine Katzung's Pharmacology Examination and Board Review, 14e, 2024; Chapter 34: Agents Used in Disorders of Coagulation
https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461&sectionid=285595695

Key concepts

Understanding platelet activation, the coagulation cascade, including amplification of thrombin, and the fibrinolytic process, is essential for understanding the pharmacology.

- Anticoagulants, fibrinolytics, and antiplatelet drugs act at specific points in the clotting cascade.
- The actions determine their therapeutic effects and adverse effects.
- The Notes Handout for this lecture includes detailed explanation of the cascades of reactions, defines the methods for monitoring activity, and the reversal agents. I also recommend watching a video on YouTube or other site.

Terms

aPTT	Activated partial thromboplastin time is for monitoring UFH and direct thrombin inhibitor therapy.
Anti-factor Xa assay	Chromogenic assay for monitoring the activity of factor Xa inhibitors
ACT	Activated clotting time, a rapid point-of-care test to monitor the activity of high-dose heparin and direct thrombin inhibitors
PT/INR	Prothrombin time is the most commonly used test for measuring warfarin therapy. International normalized ratio is a calculation for standardizing prothrombin time.
AT	Antithrombin

UFH	Unfractionated heparin
LMWH	Low molecular weigh heparin
HIT	Heparin-induced thrombocytopenia
DOAC	Direct oral anticoagulant
DTI	Direct thrombin inhibitor
DVT	Deep vein thrombosis
PE	Pulmonary embolism
VTE	Venous thromboembolism
STEMI	ST-elevation myocardial infarction
PCI	Percutaneous coronary intervention
tPA	Tissue plasminogen activator
PCC	Prothrombin complex concentrate (a human plasma product)

Antithrombotic drugs are used to prevent or treat blood clots. They are broadly categorized into three main types: **anticoagulants**, **antiplatelet drugs**, and **thrombolytics**.

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Anticoagulants	Fibrinolytics	Antiplatelet Drugs
Warfarin	Thrombolytics	Antiplatelet Drugs
Heparin, unfractionated (UFH)	Alteplase	Aspirin
LMWHs	Reteplase	Clopidogrel
Enoxaparin	Tenecteplase (TNKase)	Prasugrel
Dalteparin	Streptokinase	Ticagrelor
Fondiparinux	Urokinase	Dipyridamole
Direct Thrombin Inhibitors		Cilostazol
Bivalirudin		
DOACs		
Factor Xa Inhibitors:		
Rivaroxaban (Xarelto),		
Apixaban (Eliquis),		
Edoxaban (Savaysa)		
Direct thrombin inhibitor		
Dabigatran (Pradaxa)		

Anticoagulants inhibit either the action of coagulation factors or prevent their synthesis.

- Heparin is an injectable, rapid-acting drug with an indirect anticoagulant action by accelerating the action of antithrombin (AT) by ~1000-fold.
  - Heparin has no intrinsic anticoagulation activity.
  - AT is a small glycoprotein that inhibits serine proteases of the contact activation pathway.
- Unfractionated heparin (UFH) is a natural product extracted from porcine intestinal mucosa. UFH is mixture of straight-chain, anionic glycosaminoglycans with a wide range of molecular weights.
- UFH reversibly binds with high affinity to a specific pentasaccharide sequence on AT, causing a conformational change that opens AT's catalytic site. Thrombin must also bind heparin at a site near the pentasaccharide site. AT inactivates thrombin, factor Xa, and other coagulation factors.

Anticoagulants: UFH

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- Heparin prevents the progression of existing clots by preventing further clot formation. It has many clinical applications. UFH is administered as an IV loading dose followed by continuous infusion or subQ administration (prophylaxis use only). It binds many plasma proteins, neutralizing its activity and causing unpredictable pharmacokinetics. It is eliminated by the reticuloendothelial system (nonrenal elimination). UFH has a half-life of 1-2 hours after discontinuation. UFH therapy is monitored by aPTT, anti-factor Xa assay, and ACT.
- UFH may cause heparin induced thrombocytopenia – HIT. HIT type I is a mild reduction in platelets during the first two days of exposure, which spontaneously returns to normal and is not associated with thrombosis.
- HIT type II is an immune-mediated potentially life-threatening complication of heparin therapy when IgG attached to the heparin-PF4 complex binds the Fc receptor on the platelet surface, leading to platelet activation and a severe hypercoabulable state. Heparin must be discontinued and anticoagulation with a non-heparin anticoagulant initiated. HIT type II occurs usually 5 to 14 days after initiation of therapy.

Anticoagulants LMWH

- Low-molecular-weight forms of heparin (LMWHs), enoxaparin, dalteparin, and tinzaparin, complex with antithrombin at the pentasaccharide site, accelerating AT's inactivation of factor Xa. They have low antithrombin activity. Because of their shorter molecular sequence, they have low affinity for thrombin.
- LMWHs have several therapeutic uses including treatment and prevention of deep vein thrombosis (DVT), acute coronary syndromes, anticoagulation bridging, and adjunct to PCI. UFH and LMWHs are preferred anticoagulant therapy for pregnant patients.
- LMWHs are administered subcutaneously, have reliable absorption, renal excretion (dose adjustment for renal insufficiency), and biological half-life of ~4-6 hours. Onset of action is 20-30 min with full effect in about 4 hours. Monitoring is usually not required because of their predictable pharmacokinetics. If clinically appropriate, anti-factor Xa assay is used.
- There is a lower risk of HIT with LMWH use. Patients with HIT may have cross-sensitivity. LMWHs should not be used in patients with HIT.
- **Bleeding** is the chief adverse effect of the heparins – indeed, all anticoagulants and antiplatelet agents. Patients on anticoagulation or antiplatelet therapy are at increased risk of complications of **neuraxial anesthesia** techniques, esp. spinal epidural hematoma.
- Protamine sulfate is a specific antidote for reversal of heparin. It directly binds UFH, neutralizing it activity. LMWHs are less easily inactivated by protamine.

Anticoagulants: Fondaparinux

- Fondaparinux is a synthetic pentasaccharide that reversibly binds the pentasaccharide site on AT, leading to selective inactivation of factor Xa.
- Fondaparinux therapeutic uses include treatment and prophylaxis of DVT, PE, and prophylaxis in patients with HIT.
- It is administered subcutaneously, has reliable absorption, renal excretion (dose adjustment for renal insufficiency), and half-life of ~17 hours (1x daily administration).
- Monitoring is not required due to its predictable pharmacokinetics. If clinically appropriate, monitoring is by anti-factor Xa assay.
- Fondaparinux is not inactivated by protamine.

Direct-acting Oral Anticoagulants (DOACs): Factor Xa inhibitors

- The DOAC factor Xa inhibitors, rivaroxaban, apixaban, and edoxaban, bind selectively and reversibly to factor Xa, reducing production of thrombin from prothrombin.
- They are used for the treatment and prophylaxis of DVT and PE, post-surgical thromboprophylaxis, and non-valvular atrial fibrillation for prevention of stroke and systemic embolism. They should not be used for anticoagulation in patients with mechanical heart valves due to lack of evidence of safety and efficacy.
- Rivaroxaban and apixaban are metabolized by CYPs including CYP3A4. ***Strong CYP inhibitors and inducers should be avoided.*** Edoxaban is excreted mainly unchanged in the urine. All are substrates of P-glycoprotein. ***Strong P-glycoprotein inhibitors and inducers should be avoided.***
- Monitoring is generally not required. If clinically appropriate, anti-factor Xa assay is used.
- Bleeding is the most serious adverse effect. There is no specific antidote. PCC is used for reversal in patients with severe bleeding.

Direct-acting Oral Anticoagulants (DOACs): Direct thrombin inhibitor

- Dabigatran is an oral DTI included in the DOACs group. It inhibits both free and clot-bound thrombin by binding specifically and reversibly to the catalytic site of thrombin.
- Like the other DOACs, it is used for the treatment and prophylaxis of DVT and PE, VTE, and non-valvular atrial fibrillation, and may be used in patients with HIT. It is contraindicated in patients with mechanical heart valves and not recommended in those with bioprosthetic heart valves. Dabigatran demonstrated inferior efficacy and higher risk of thromboembolic events and bleeding compared to warfarin.
- Dabigatran is available as he prodrug dabigatran etexilate, which is hydrolyzed by plasma esterases to active dabigatran and excreted in urine. It is a P-glycoprotein substrate and ***P-glycoprotein strong inducers and inhibitors should be avoided***. Therapeutic monitoring is not required.
- Idarucizumab is a dabigatran-specific reversal agent that neutralizes its anticoagulant action.

Parenteral Direct Thrombin Inhibitors

- Argatroban is a synthetic intravenous anticoagulant derived from L-arginine that reversibly binds thrombin's catalytic site. Its effects are immediate. It is used for the prophylaxis and treatment of thromboembolism in patients with HIT, and during PCI in patients who have or are at risk for developing HIT. Argatroban is metabolized in the liver and has a half-life of <1 hour.
- Bivalirudin is an analog of hirudin, a naturally occurring peptide in leech salivary glands. Bivalirudin has a bifunctional action; it specifically and reversibly binds to thrombin's catalytic site and exosite 1. It is used for anticoagulation in patients with HIT, undergoing PCI, and cardiac surgery as an alternative to heparin. Bivalirudin is metabolized by proteolysis and has a half-life of about 25 minutes (normal renal function).
- Anticoagulation monitoring is done using aPTT or ACT.
- PCC is used for reversal of severe bleeding.

Anticoagulants: Vitamin K Antagonists

Warfarin is the only coumarin anticoagulant available in the US.

- Factors II (prothrombin), VII, IX, and X, which are synthesized by the liver, require vitamin K as a cofactor in the glutamation of a number of their residues by gamma-carboxyglutamase. The gamma-carboxyglutamyl residues bind calcium ions, which are essential for interaction between the coagulation factors and platelet membranes.
- In the carboxylation reactions, the reduced vitamin K cofactor is converted to vitamin K epoxide during the reaction. Vitamin K is regenerated from the epoxide by vitamin K epoxide reductase (VKOR1C), the enzyme that is inhibited by warfarin, which reduces the activity of these clotting factors by up to 40%. Anticoagulant effects are delayed for 72 to 96 hours – the time required to deplete the pool of circulation clotting factors.
- Therefore, bridging therapy with a rapid acting anticoagulant (UFH, LMWH, or DOAC) may be required.

Warfarin

- Warfarin has many therapeutic uses, such as prevention of thromboembolism in patients with atrial fibrillation with or without prosthetic heart valves, treatment and prophylaxis of DVT and PE, and others.
- Warfarin is available as a racemic mixture taken orally with rapid complete absorption. It is highly bound to plasma proteins. Warfarin crosses the placenta. The active S-isomer is metabolized mainly by CYP2C9 to inactive metabolites, which are excreted in urine and feces. Numerous drug-food and drug-drug interactions may increase or decrease warfarin’s effect.
- Bleeding is the major adverse effect. Vitamin K administration orally or IV is used to reverse bleeding. Replacement of coagulation factors with PCC or fresh frozen plasma may be need for severe bleeding.
- Rare complications are skin lesions and necrosis and cholesterol emboli causing purple (blue) toe syndrome.
- Warfarin is teratogenic and contraindicated in pregnancy.

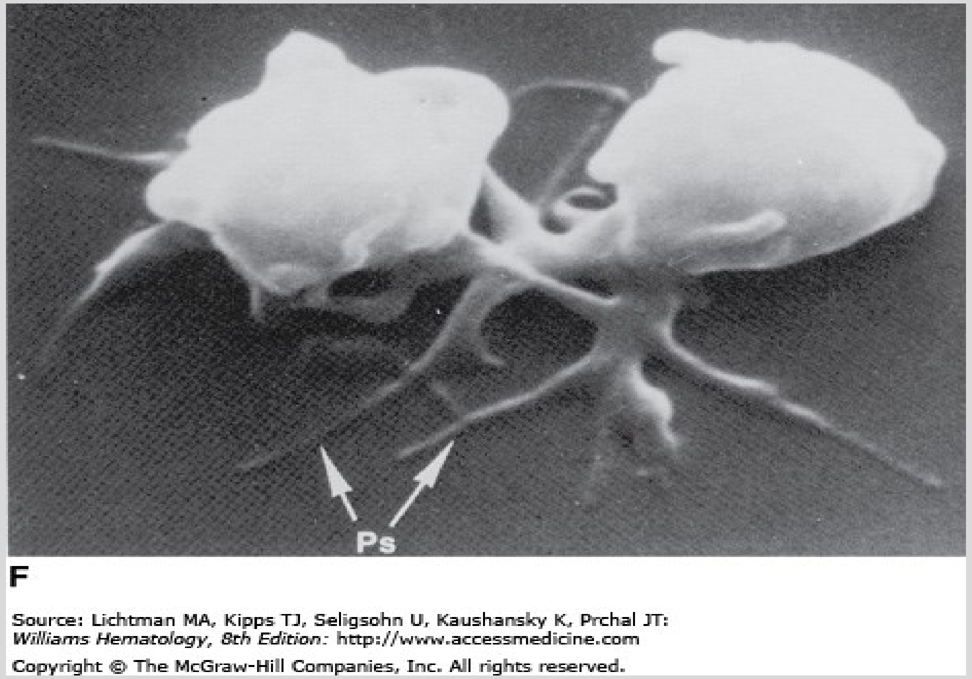
Brief Overview of Blood Coagulation

Recommendation: Watch a teaching video explaining the steps and amplification of the clotting cascade. Here is a suggestion:

Khan Academy video created by Patrick van Nieuwenhuizen

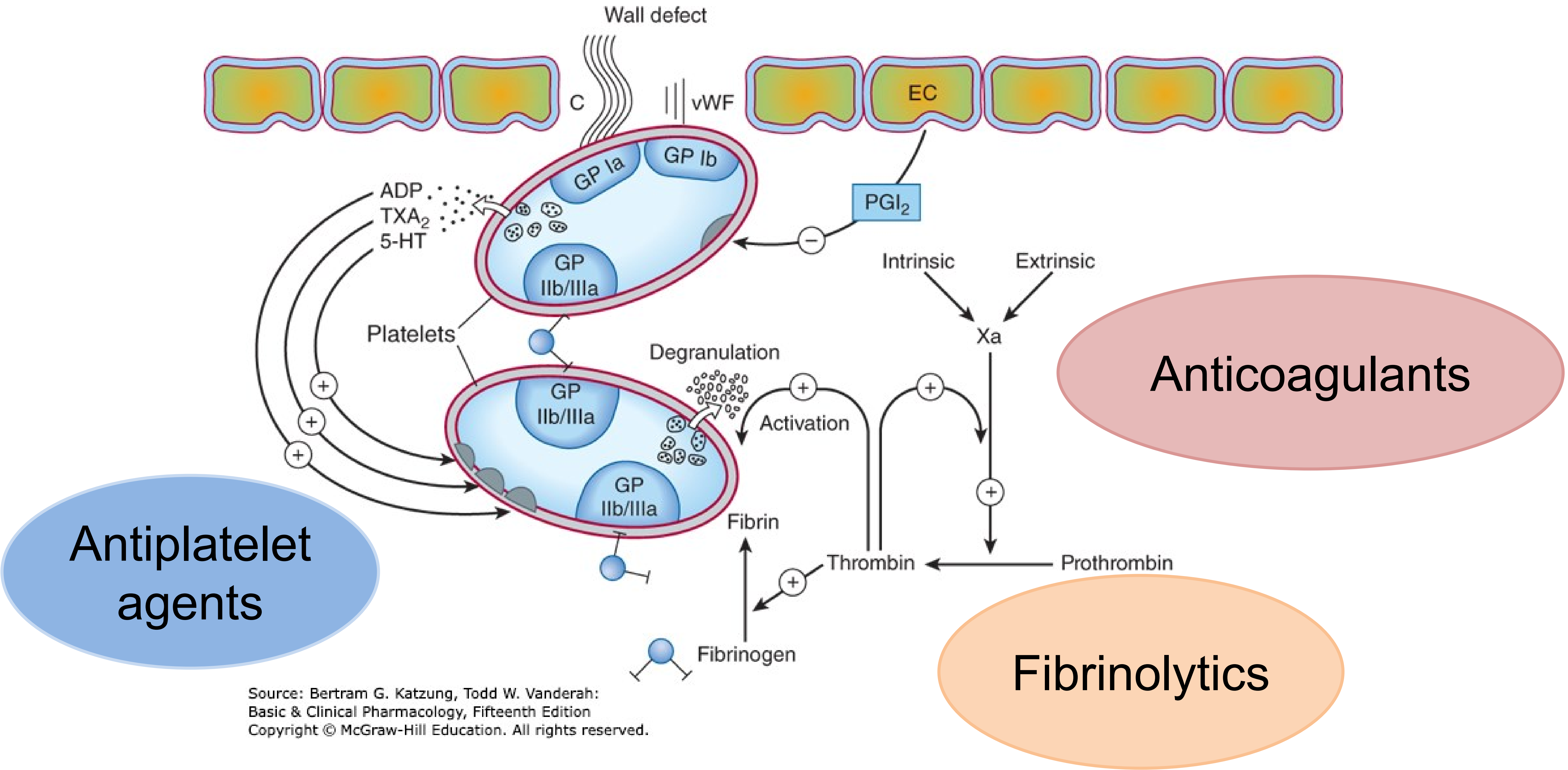
https://www.khanacademy.org/test-prep/mcat/organ-systems/hematologic-system/v/coagulation-cascade

Hemostasis:
Cessation of blood loss from a damaged vessel



Williams Hematology 9e Figure 112-2F
Activated platelets with filopodia (pseudopodia)

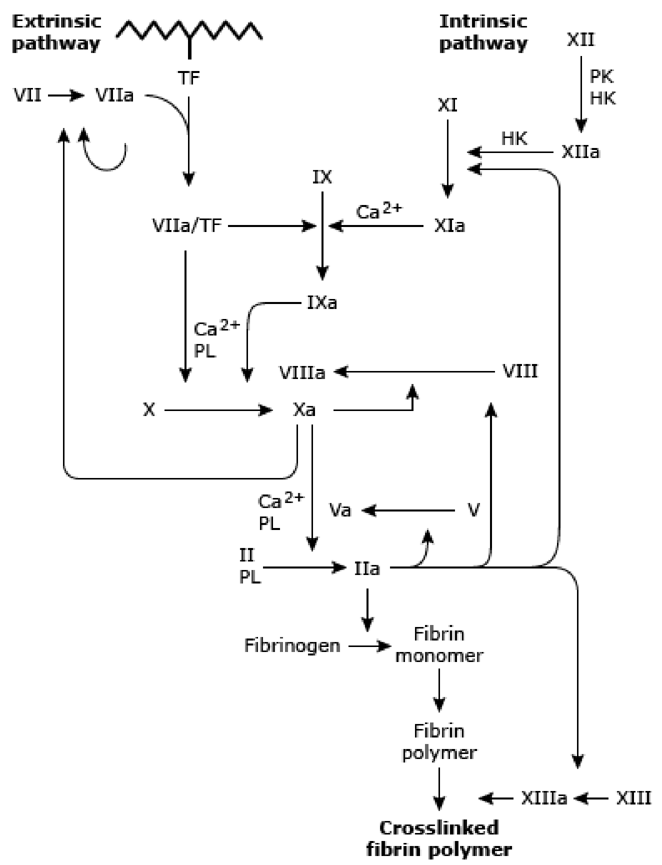
1. Platelet adhesion
2. Platelet activation
3. Platelet aggregation
4. Activation of plasma coagulation factors
5. Generation of fibrin clot, which reinforces platelet aggregate
6. Wound healing follows: Degradation of platelet aggregate and fibrin clot



Thrombus formation at the site of the damaged vascular wall (EC, endothelial cell) and the role of platelets and clotting factors. Platelet membrane receptors include the glycoprotein (GP) Ia receptor, binding to collagen (C); GP Ib receptor, binding von Willebrand factor (vWF); and GP IIb/IIIa, which binds fibrinogen and other macromolecules. Antiplatelet prostacyclin (PGI_2) is released from the endothelium. Aggregating substances released from the degranulating platelet include adenosine diphosphate (ADP), thromboxane A_2 (TXA_2), and serotonin (5-HT). Production of factor Xa by intrinsic and extrinsic pathways is detailed in Figure 34–2.(Labels in circles are mine – L. Goldstein)

Citation: Chapter 34 Drugs Used in Disorders of Coagulation, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021.
Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250599717 Accessed: April 20, 2021

Coagulation cascade detailed/traditional view



Schematic representation of the coagulation cascade including our improved understanding of the role of the tissue factor (TF) pathway in initiating clotting, interactions between pathways, and the role of thrombin in sustaining the cascade by feedback activation of coagulation factors. Factor II is prothrombin, and factor IIa is thrombin.

PK: prekallikrein; HK: high molecular weight kininogen; PL: phospholipid.

Adapted from: Ferguson JD, Banning AP. Spontaneous intramural aortic haematoma: incidence, prognosis and complications. Eur Heart J 1998; Suppl 19:8.

UpToDate®

Two sequences can precipitate coagulation: the extrinsic pathway – tissue factor activation pathway – and the intrinsic pathway – surface contact activation pathway:

Tissue factor activation pathway The primary physiologic event initiating clotting

1- Endothelial damage → expression of TF

2- FVIIa + TF + Ca^{2+} complex (extrinsic ten-ase) activates fX and fIX → FXa and FIXa

The extrinsic pathway converges with the common pathway.

Surface contact activation pathway:

1,2- fXII contact with negative surface → FXIIa → FXIIa with HMWK cleaves XI → FXIa.

3- FXIa activates factor IX → FIXa.

❖ ***Thrombin*** amplification: A small initial amount of thrombin generated from interactions between TF + FVIIa activates factor XI → FXIa → ***amplification of thrombin*** generation.

4- FIXa + FVIIIa + Ca^{2+} complex + phospholipids (negative charge) form intrinsic ten-ase.

Ten-ase converts factor X → FXa.

5- FXa activates fVIII → FVIIIa and thrombin activate fVIII → FVIIIa

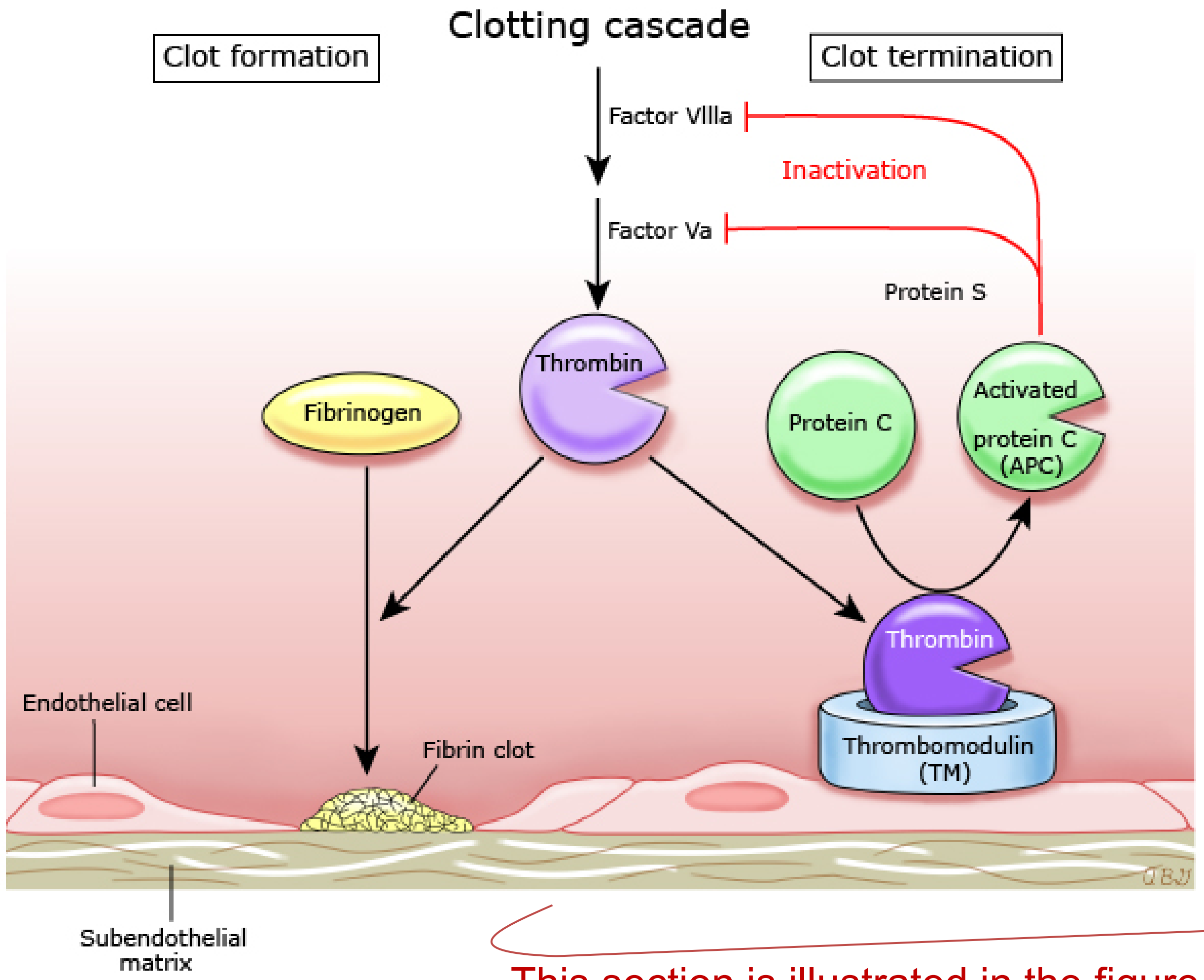
- progressively increasing fVIII activation as FXa and thrombin are formed.

The intrinsic pathway converges with the common pathway.

Common Pathway

- FXa + FVa (cofactor) + phospholipid → prothromibinase
- Prothrombinase cleaves prothrombin (factor II) → ***thrombin*** (FIIa).
- Thrombin converts fibrinogen (soluble plasma protein) → fibrin clot (insoluble)

Hemostatic control mechanisms and termination of clotting



This section is illustrated in the figure.

Refer to UpToDate for an overview of hemostasis regulation and the roles of specific procoagulant and anticoagulant factors in clinical thrombosis and hemostasis.

APC: activated protein C; TM: thrombomodulin.

Courtesy of Lawrence LK Leung, MD

UpToDate®

NATURAL ANTICOAGULANT MECHANISMS

Regulators of blood coagulation

- **Tissue factor pathway inhibitor (TFPI):** A plasma protein that binds to the TF-VIIa after activation of factor X to Xa.
- **Antithrombin (AT):** A small plasma glycoprotein synthesized in the liver and circulating in plasma that inactivates several enzymes of the coagulation system, particularly thrombin and factor Xa (also factors IXa, XIa, XIIa).

Pharmacology relevance: Heparin sulfate drugs, like endogenous glycosaminoglycans, amplify AT action ~1000-fold.

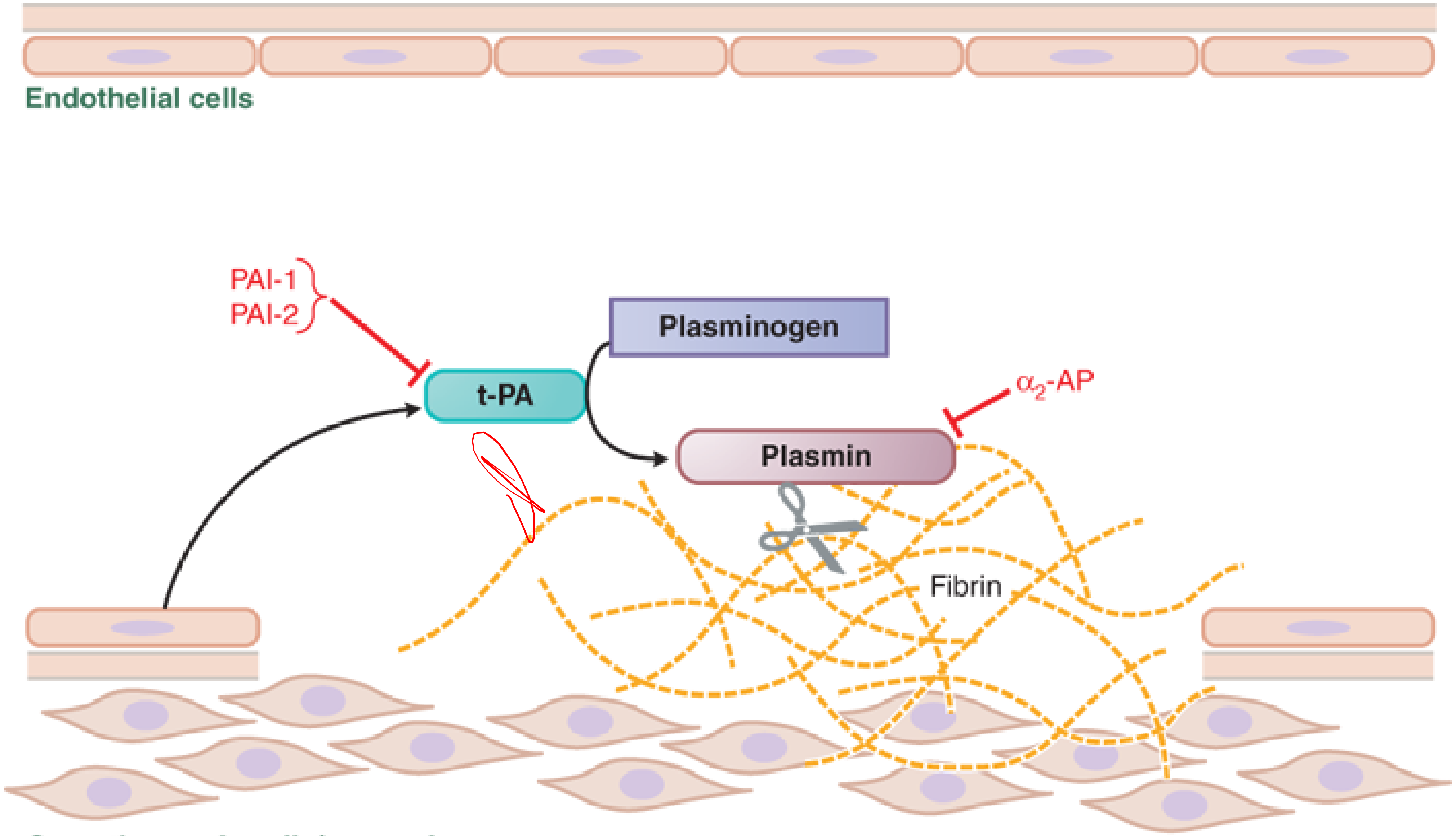
- **Thrombomodulin:** A high affinity thrombin receptor on the endothelial cell membrane that forms a complex with thrombin and removes thrombin from circulation.
- **Protein C:** A vitamin K-dependent plasma zymogen requires activation by thrombomodulin-bound thrombin.

Activated protein C is a protease that, with cofactor protein S, inactivates the FVa and FVIIIa. It also inhibits plasminogen activator inhibitor-1 (PAI-1) and PAI-2, thereby facilitating fibrinolysis.

- **Protein S:** Vitamin K-dependent glycoprotein; Cofactor of protein C
- **Nitric oxide (NO)** and **prostacyclin (PGI_2):** Synthesized by endothelial cells and released into the blood induce vasodilation and inhibit platelet activation and subsequent aggregation.

Fibrinolysis

- Endothelial cells secrete tissue plasminogen activator (tPA) at sites of injury.
- tPA binds to fibrin and converts plasminogen to plasmin, which digests fibrin.
- Plasminogen activator inhibitors-1 and -2 (PAI-1, PAI-2) inactivate tPA
- Antiplasmin (a_2-AP) inactivates plasmin.



Source: Laurence L. Brunton, Björn C. Knollmann:
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:

Citation: Chapter 36 Blood Coagulation and Anticoagulant, Fibrinolytic, and Antiplatelet Drugs, Brunton LL, Knollmann BC. *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e;* 2023, figure 36-4. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=3191&sectionid=266700467 Accessed: March 30, 2023

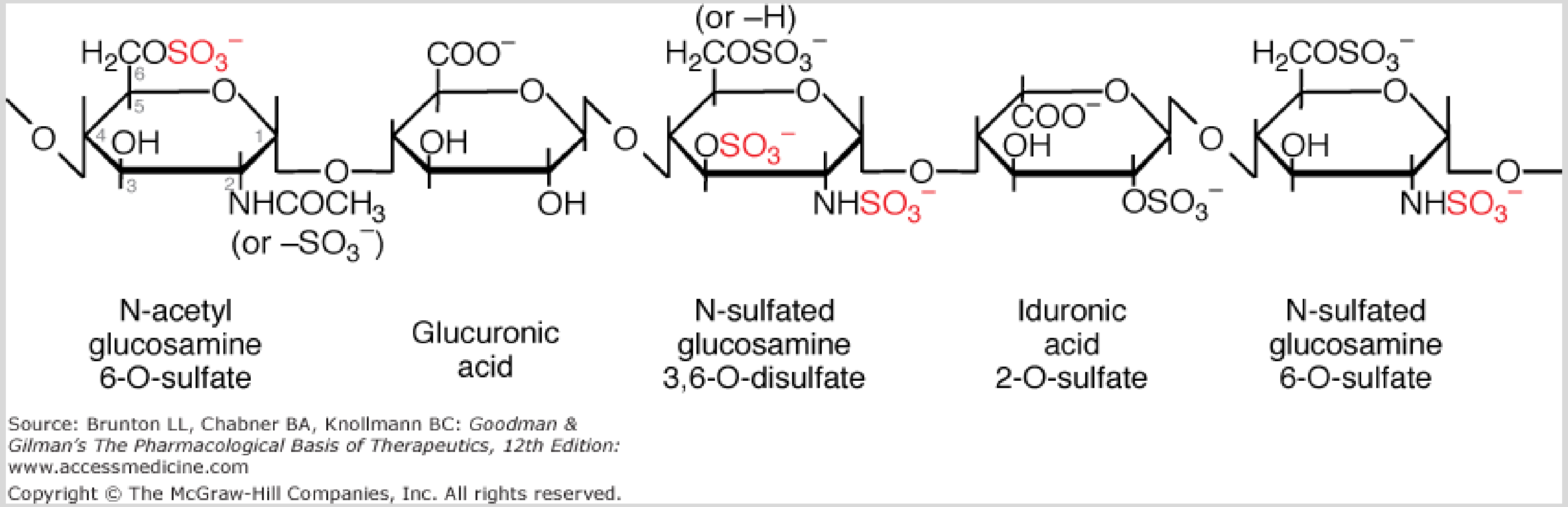
Slide adapted by LG

Anticoagulants

Unfractionated heparin (UFH)
Low molecular weight heparin (LMWH)
Fondaparinux
Direct-acting factor Xa Inhibitors
Direct thrombin inhibitors (DTIs)

unfractionated

✳Heparin



Sulfated anionic linear polysaccharides (heterogeneous mixture)
Molecular weight: 3,000-30,000 Da (mean ~15,000 Da)

Heparin chains of appropriate length bind heparin's target, antithrombin.
Heparin chains also bind to endothelial cell surfaces and various plasma proteins, neutralizing heparin's action.

Heparin (UFH) Pharmacokinetics:

<table>
<tr><td>A</td><td colspan="2">I.V. infusion: onset within minutes
subQ: erratic absorption (only for prophylaxis) | onset 20-30 minutes</td></tr>
<tr><td>D</td><td>– Nonspecific binding endothelial cells and plasma proteins
– Polarity and large size prohibits crossing of cell membranes</td><td>– Does not cross blood-brain barrier
– Does not cross the placenta
– Does not enter breast milk</td></tr>
<tr><td>E</td><td colspan="2">Nonrenal elimination – Eliminated by the reticuloendothelial system (mainly liver and spleen)</td></tr>
<tr><td>$t_{1/2}$</td><td colspan="2">Mean 1.5 h (adults) after completion of infusion
Half-life is affected by hepatic cirrhosis, obesity, malignancy, presence of pulmonary embolism, and infections</td></tr>
</table>

Monitoring UFH anticoagulation: aPTT, anti-factor Xa assay, ACT
Because of protein and tissue binding, free heparin levels are variable after administration

Monitor platelet counts due to potential for heparin-induced thrombocytopenia.

Heparin is a catalyst that enhances the inhibitory activity of antithrombin against serine proteases.

Heparin reversibly binds AT at its pentasaccharide sequence

↓

causes a conformational change

that reveals AT's reactive center loop

—UFH must simultaneously bind AT and thrombin

to potentiate thrombin inhibition—

↓

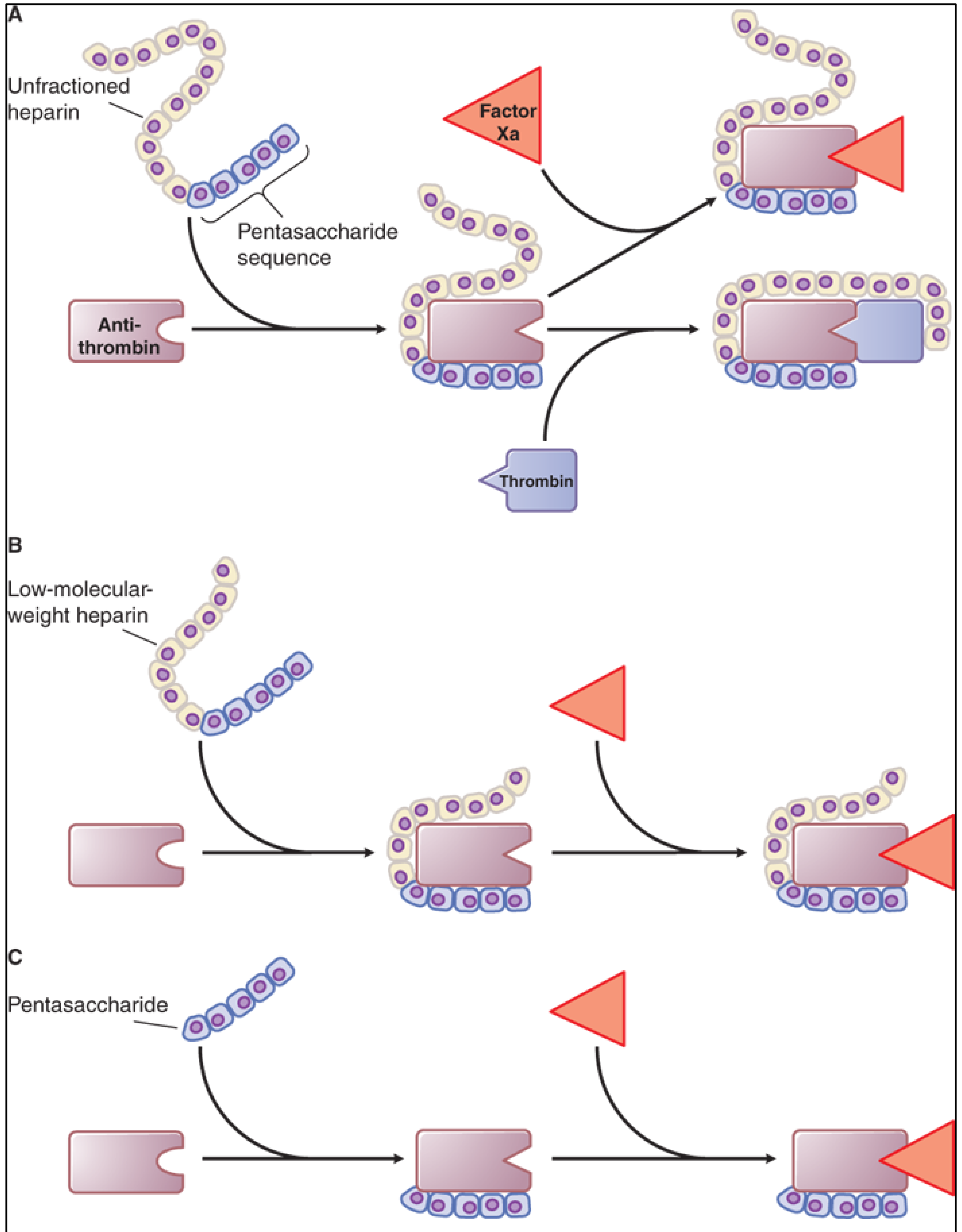
accelerates AT's protease inactivating action:

thrombin (FIIa) and FXa (and others)

~1000 times faster

↓

inactivation of the coagulation factors



LMWH and Fondaparinux accelerate AT inactivation of fXa.

❖ Note: Heparin agents have no intrinsic anticoagulant activity.

Heparin Goal of Therapy

To reduce blood coagulability to a level that is low enough to prevent thrombosis but not so low as to promote spontaneous bleeding

Heparin prevents progression of existing clots by inhibiting further clotting.

UFH Therapeutic Uses (many)	Adverse Effects
Rapid onset of action • DVT and PE treatment and prevention • Acute MI or unstable angina, initial mgmt. • Cardiopulmonary bypass surgery • Renal dialysis • Pregnancy: UHF, LMWH, or fondaparinux are the preferred agents	• Bleeding – the major adverse effect of anticoagulants and antiplatelet agents. • Neuraxial anesthesia techniques → spinal or epidural hematomas → pressure on spinal cord or cauda equina → paralysis • Heparin-induced thrombocytopenia (HIT) – Type I: Benign, 1-2 days of therapy, spontaneous return to normal – Type II: Immune mediated; life-threatening • Osteoporosis (long-term use)

Heparin resistance: Very high heparin dose is needed to achieve therapeutic aPTT or ACT

- Congenital or acquired AT deficiency
- Elevated levels of plasma proteins that bind heparin
- Elevated acute phase proteins, factor VIII and fibrinogen

Pathogenesis of immune-mediated HIT.

IgG is the autoantibody against the heparin–PF4 complex.

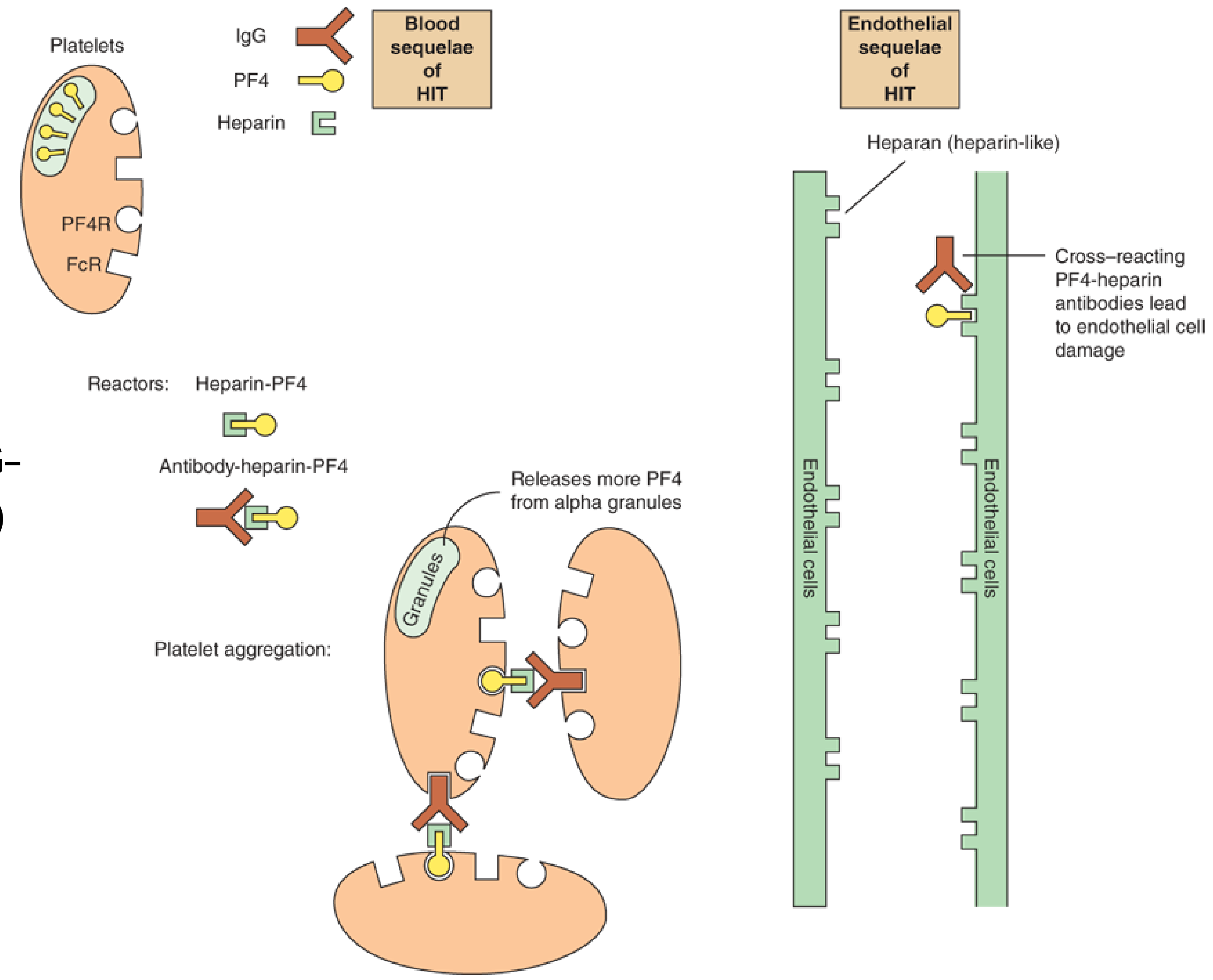
***HIT occurs if IgG attached to the heparin-PF4 complex binds the Fc receptor on the platelet surface, leading to platelet activation*.**

(Platelets can bind to each other and become activated via the IgG-Fc receptor interaction, the PF4–PF4 receptor interaction, or both.)

Activated platelets release pro-thrombotic substances and PF4. IgG activates more platelets, more PF4 is released forming more complexes with heparin, activating more platelets.

A severely hypercoagulable state results.

Thrombocytopenia results from consumption of IgG-coated platelets by cells of the reticuloendothelial system.



Source: Gary D. Hammer, Stephen J. McPhee: Pathophysiology of Disease: An Introduction to Clinical Medicine, Eighth Edition

Discontinue heparin.

Start a non-heparin anticoagulant: direct thrombin in inhibitor, fondaparinux, or DOAC.

Citation: Chapter 6 Blood Disorders, Hammer GD, McPhee SJ. *Pathophysiology of Disease: An Introduction to Clinical Medicine, 8e;* 2019, Figure 6-13. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2468&sectionid=198220627 Accessed: April 03, 2023

Slide adapted by LG

Heparin Antidote: Protamine sulfate

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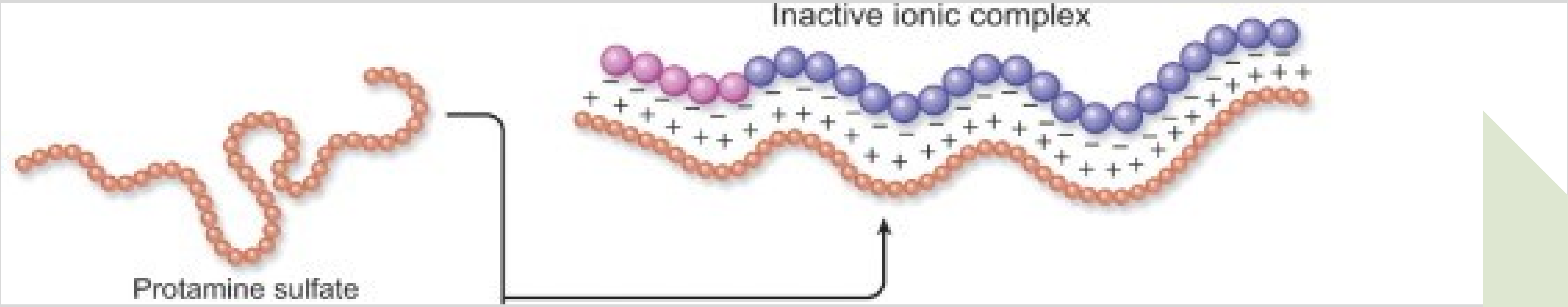
Protamine is a highly basic polypeptide derived from salmon sperm.



Protamine's multiple positively charged groups bond ionically with negatively charged groups on heparin

Neutralization begins immediately

Duration ~2 hours

The protamine-heparin complex is cleared

Caution: Excess protamine has anticoagulant effect

Figure: www.sciencedirect.com

Low molecular weight heparins (LMWH)

✳Enoxaparin

Dalteparin

Tinzaparin

Synthetic pentasaccharide

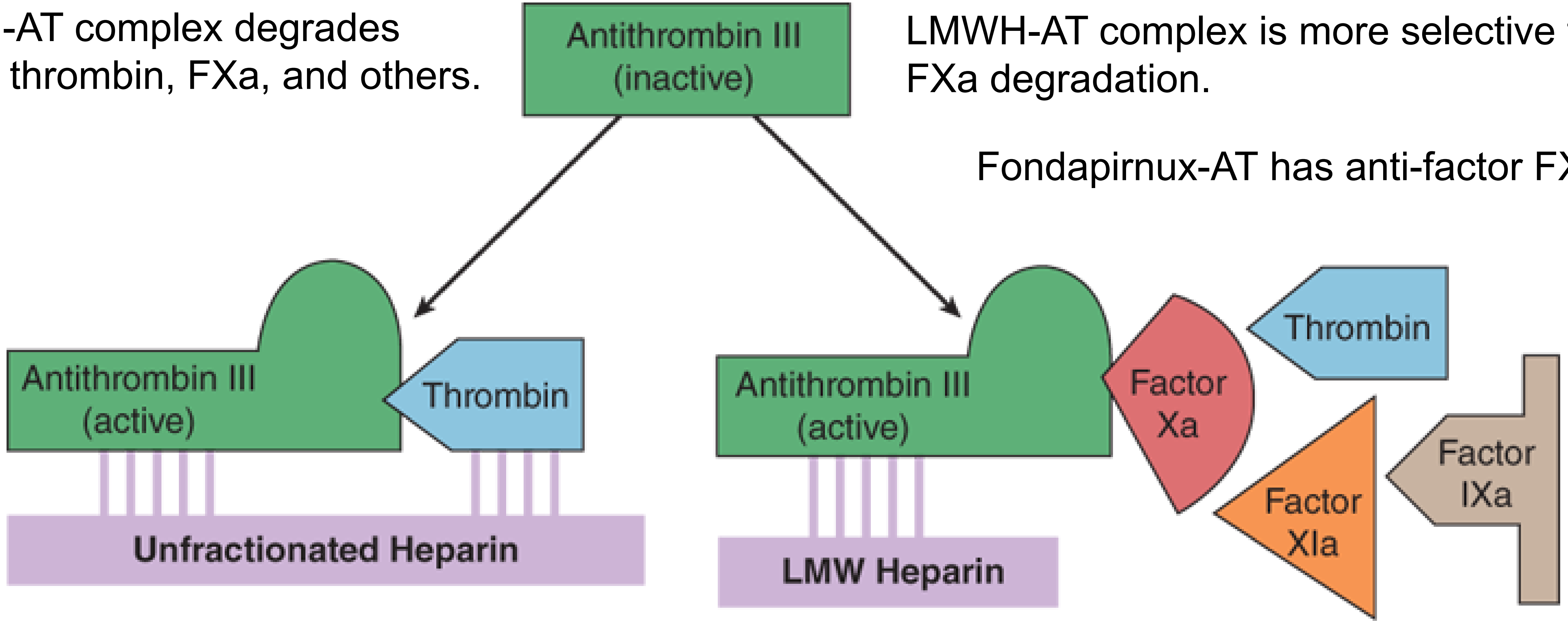
✳Fondaparinux

Comparing actions of UHF – LMWH - Fondaparinux

UFH-AT complex degrades both thrombin, FXa, and others.

LMWH-AT complex is more selective for FXa degradation.

Fondapirnux-AT has anti-factor FXa only.



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition

Differences between low-molecular-weight (LMW) heparins and high-molecular-weight heparin (unfractionated heparin). Fondaparinux is a small pentasaccharide fragment of heparin. Activated antithrombin III (AT III) degrades thrombin, factor X, and several other factors. Binding of these drugs to AT III can increase the catalytic action of AT III 1000-fold. The combination of AT III with unfractionated heparin increases degradation of both factor Xa and thrombin. Combination with fondaparinux or LMW heparin more selectively increases degradation of Xa.

Citation: Chapter 34 Drugs Used in Disorders of Coagulation, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021.
Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250599717 Accessed: April 20, 2021

LMWH / Fondaparinux PK, Therapeutic uses, and Cautions

LMWHs	Fondaparinux
SubQ; reliable absorption; renal excretion; biological t½ 4-5 hours	SubQ; reliable absorption; renal excretion; biological t½ ~17 hours
Use: • DVT / PE treatment and prophylaxis • Acute coronary syndromes • Warfarin bridging • PCI adjunctive use • Cautions: as for unfractionated heparin	Uses: • DVT / PE treatment and prophylaxis • HIT anticoagulation • Acute thrombosis unrelated to HIT in patients with a history of HIT • Acute symptomatic superficial vein thrombosis (≥5 cm in length) of the legs

Routine anticoagulation monitoring usually not required. Anti-factor Xa-assay if clinically indicated.

Adverse Effects • Immune-mediated HIT occurs less frequently with LMWHs • Cross-sensitivity may occur • Avoid in patients with HIT	• Adverse effects • Avoid fondaparinux alone in PCI due to guided catheter thrombosis. The fondaparinux molecule is too short to block the thrombotic process.

Common: Bleeding; Avoid in neuraxial anesthesia.

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Comparison of heparin, low molecular weight heparin, and fondaparinux

FEATURES	HEPARIN, UHF	LMWH	FONDAPARINUX
Source	Biological	Biological	Synthetic pentasaccharide
Mean molecular weight	15,000 Da	5,000 Da	1,500 Da
Target	Xa and IIa	Xa	Xa
Bioavailability (subQ)	30 %	90 %	100 %
$t_{1/2}$	1 hour	4 hours	17 hours
Renal excretion	No	Yes	Yes
Antidote effect (protamine)	Complete	Partial	None
Thrombocytopenia	<5%	<1%	Does not cause thrombocytopenia

Da: daltons

Adapted from Goodman & Gilman, 2023 Table 36-1

Direct oral anticoagulants – DOACs:

Direct Factor Xa Inhibitors

✳Rivaroxaban

Apixaban

Edoxaban

Pharmacokinetics of Direct Factor Xa Inhibitors

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	Rivaroxaban	Apixaban	Edoxaban
A	Oral	Oral	Oral
D	Protein binding >90% P-glycoprotein substrate	Protein binding ~90% P-glycoprotein substrate	Protein binding ~50% P-glycoprotein substrate
M	Hepatic CYP3A4/5 Bioavailability 66-100% dose-dependent	CYP3A4/5 predominantly Bioavailability 50%	Minimal metabolism Bioavailability ~60%
E	Renal, ~35% unchanged and fecal excretion	Renal, ~30 % unchanged and fecal excretion	Urine unchanged
t½	5-9 hours 11-13 hours in elderly 2x daily treatment, 1x daily prophylaxis	~8 hours (lower dose) ~15 hours (higher dose) 2x daily dose for most indications	10-14 hours once daily dosing
	Avoid use or adjust dose in patients with impaired renal function		

Direct-acting Factor Xa Inhibitors

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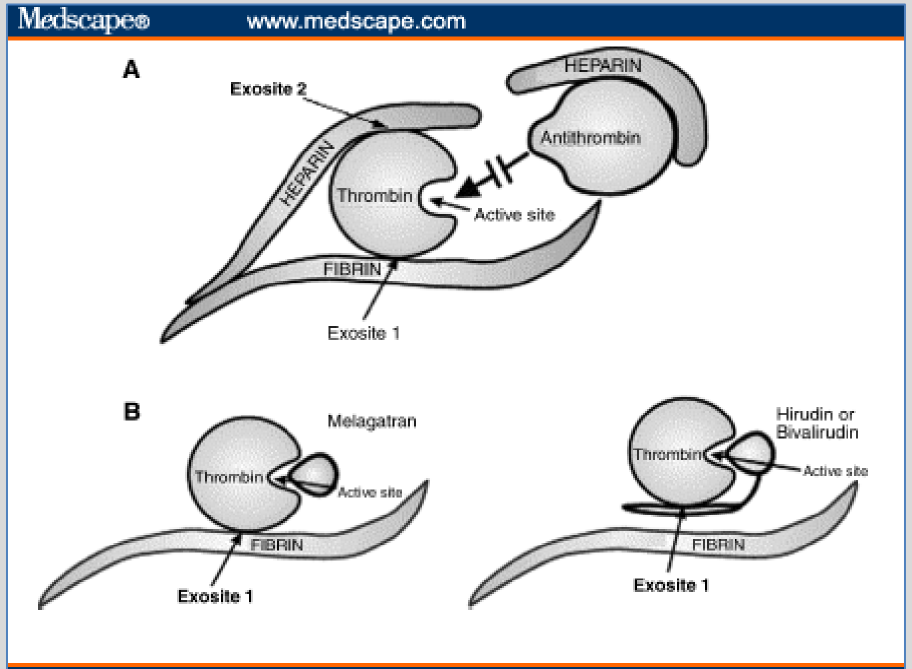
Mechanism of action	Direct, selective and reversible inhibition of free and clot-bound factor Xa, independent of AT.
Therapeutic uses	DVT and PE treatment and prophylaxis Post-operative (knee, hip) thromboprophylaxis Atrial fibrillation, nonvalvular, prevention of stroke and systemic embolus
Routine anticoagulation monitoring usually not required. Anti-factor Xa if clinically indicated.	
Adverse effects	Bleeding Neuraxial anesthesia hematoma risk
Reversal Update: Andexanet-alfa was recently withdrawn from U.S. market. Risks of thromboembolic events outweigh the benefits.	Recombinant inactivated FXa: Andexanet-alfa binds and sequesters FXa inhibitors rivaroxaban and apixaban ➢ Caution: Andexanet-alfa also inhibits tissue factor pathway inhibitor, increasing tissue factor-initiated thrombin generation. Serious and life-threatening arterial and venous thrombotic, ischemic, and cardiac events have been observed 3 to 30 days after use.
Drug interactions	Other drugs that affect coagulation and platelet function CYP3A4 and P-glycoprotein inhibitors and inducers

Direct Thrombin Inhibitors (DTIs)

DOAC	Parenteral	
∗Dabigatran	Hirudin derivatives	Synthetic
	Hirudin	∗Argatroban
	Lepirudin	∗Bivalirudin
	Desirudin	



Source: Am J Health-Syst Pharm © 2005 American Society of Health-System Pharmacists

DTIs act on free and clot-bound thrombin.

DTIs bind catalytic site of thrombin →

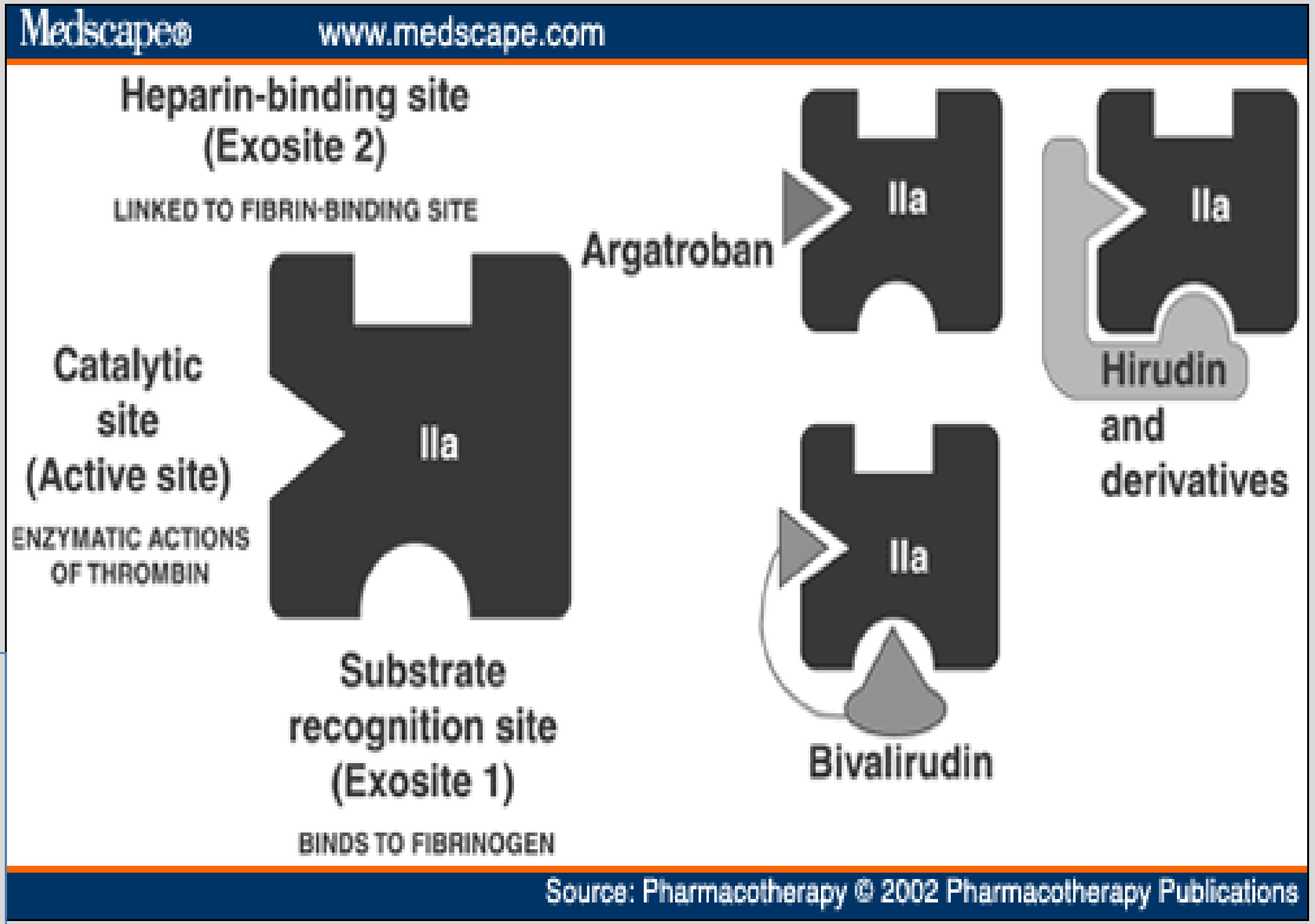
- Prevent thrombin-mediated cleavage of fibrinogen to fibrin monomers,

and

- Prevent activation of factors V, VIII, and XIII and Protein C
- Prevent platelet aggregation

Dabigatran and Argatroban reversibly and specifically bind thrombin's catalytic site.

Bivalirudin bind reversibly and specifically to the catalytic site and exosite 1.



Dabigatran etexilate: Oral Direct Thrombin Inhibitor

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PK	Etexilate oral prodrug rapidly converted to active dabigatran Moderate protein binding; P-glycoprotein substrate Rapid complete hydrolysis of etexilate $\xrightarrow{\text{plasma and hepatic esterases}}$ dabigatran $\xrightarrow{\text{hepatic glucuronidation}}$ active acylglucuronide isomers | Bioavailability 3-7% Renal excretion of dabigatran acylglucouronides | t½ ~16 hours	
MOA	Reversible selective inhibition of catalytic site of free and clot-bound thrombin	
Uses	DVT and PE prevention and treatment Atrial fibrillation, nonvalvular for prevention of thrombotic complications Contraindicated in patients with mechanical heart valves; not recommended in patients with bioprothetic valves – reduced efficacy, excess incidence of major bleeds	
Monitoring	Routine monitoring not required. Test with aPTT if clinically indicated.	
Reversal	Idarucizumab binds specifically to dabigatran and glucuronide metabolites.	
Cautions	Bleeding Neuraxial anesthesia hematoma risk Other drugs affecting coagulation Pgp inducers: contraindicated Pgp inhibitors: monitor therapy	Pregnancy (limited data) Patients ≥75 year – extreme caution Obesity (BMI≥40 kg/m²): Use alternative drug

Pharmacology of Argatroban and Bivalirudin

Argatroban synthetic, derived from L-arginine	**Bivalirudin** synthetic, 20 amino acid peptide
IV, immediate onset, moderate protein binding, minor hepatic metabolism (CYP3A4/5), feces unchanged and metabs., t½ 40-50 min	IV, immediate onset, binds only thrombin, proteolytic cleavage, urine (20%), t½ ~25 min
Uses: Anticoagulation patients with HIT PCI	Uses: Anticoagulation patients with HIT PCI Cardiac surgery alternative to heparin
Anticoagulation monitoring: aPTT, ACT	
Bleeding, neuraxial anesthesia hematoma risk, acute stent thrombosis within 4 hours in patients with STEMI undergoing PCI; hypersensitivity reactions reported with hirudin derivatives	

Desirudin: subQ for DVT prophylaxis following hip surgery | Not available for clinical use

VKOR1C Vitamin K Antagonists

✳Warfarin

Acenocoumarol available in Canada

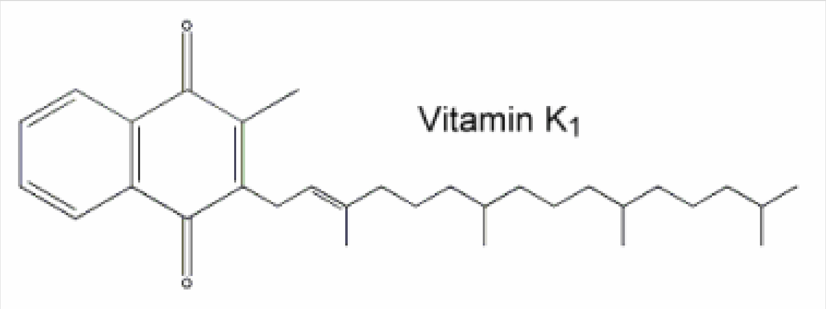
Dicoumarol and indandione derivatives withdrawn from market

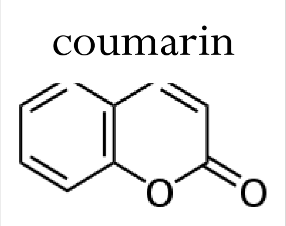
Warfarin is a racemic mixture of *S*- and *R*-warfarin: *S*-warfarin is 3-5x more potent inhibitor of vitamin K epoxide reductase.

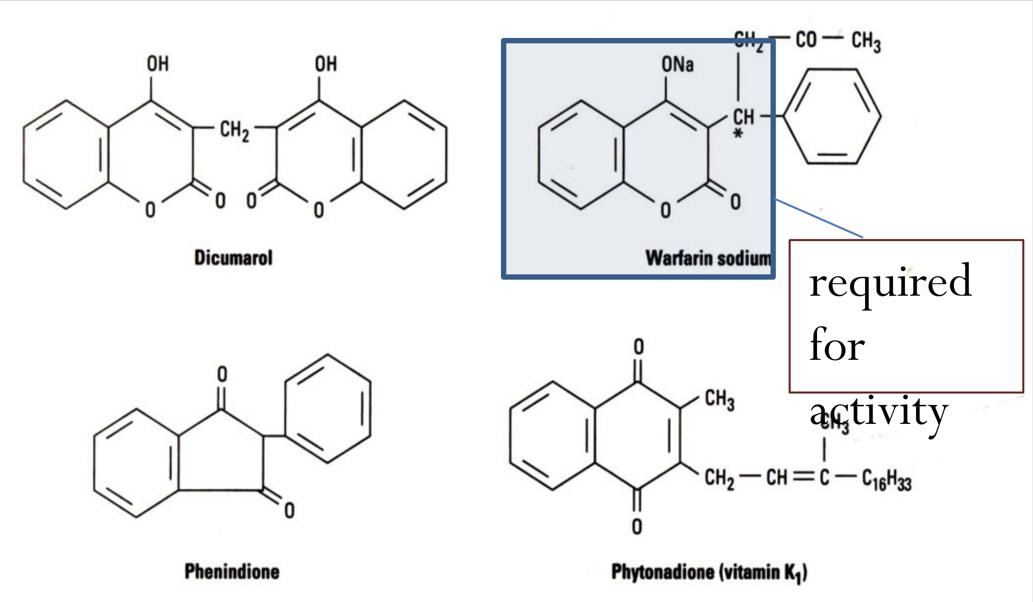
R- and *S*-enantiomers also differ in metabolism, elimination, and interactions with other drugs.

Vitamin K is a cofactor of gamma-glutamylcarboxylase in the enzymatic conversion of nonfunctional prozymogens to functional zymogens, the coagulation factors II, VII, IX, and X and anticoagulant proteins C and S.

Vitamin K is oxidized in the process and must be converted to the active reduced form by VKOR1C to continue the cycle of coagulation factor synthesis.







Warfarin Pharmacokinetics

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A	Oral, rapid complete absorption; I.V. Dose according to PT/INR results	Onset of anticoagulation: 24-72 h Full therapeutic effect 5-7 days Duration of action 2-5 days
D	>99% bound to albumin	Crosses placenta
M	Hepatic: ***S*-warfarin:** mainly **CYP2C9** | *R*-warfarin predominantly via CYP3A4 Inactive metabolites	
E	Urine, metabolites **t½** 20-60 hours (highly variable)	
	Monitoring: PT: prothrombin time and INR: international normalized ratio	
	High potential for drug-drug and drug-food (Vit K containing foods) interactions	

Factor VII	Factor IX	Factor X	Factor II	Protein C	Protein S
t½ 6 h	t½ 24 h	t½ 36 h	t½ 50 h	t½ 8 h	t½ 30 h

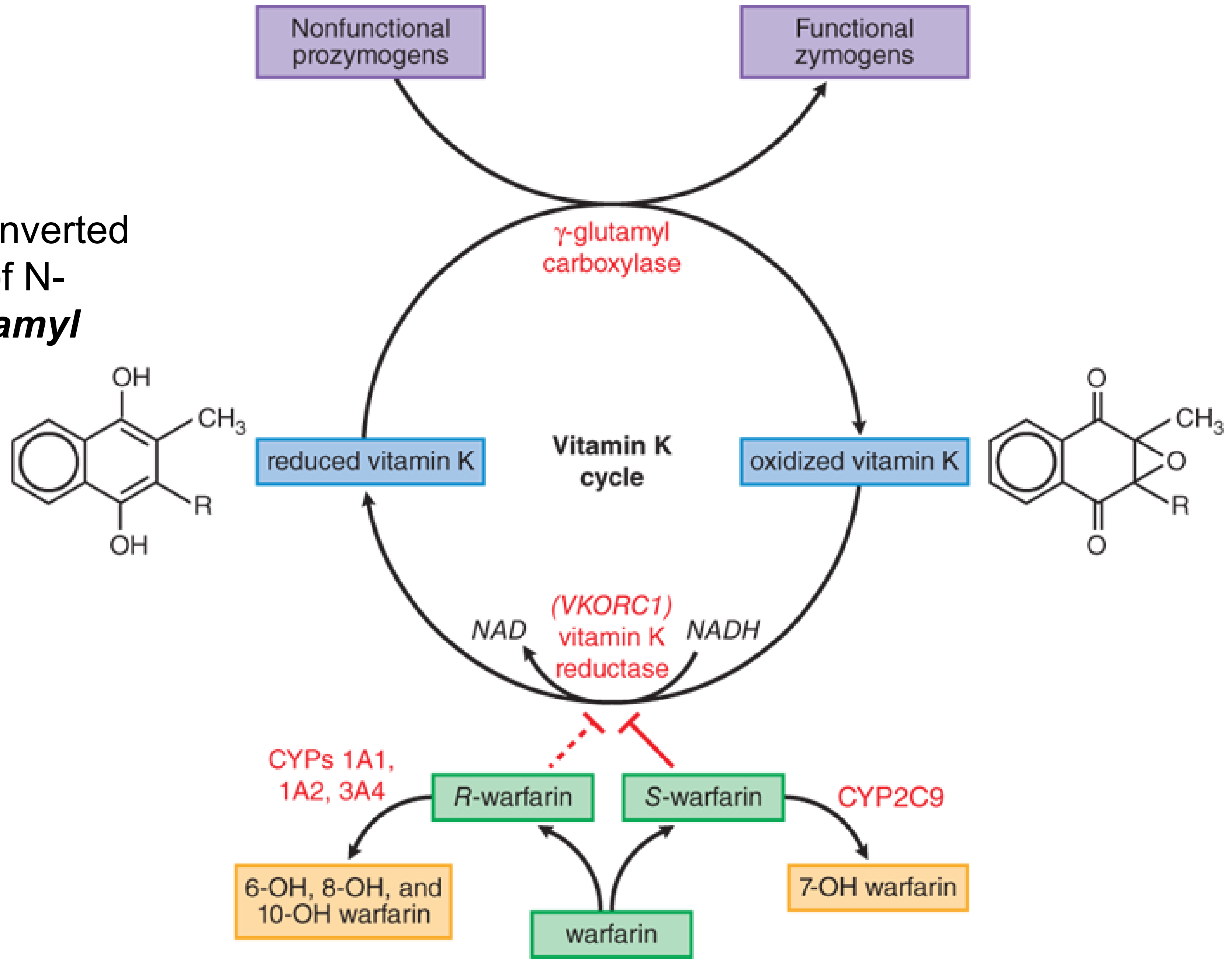
Because of the long $t_{1/2}$ of some of the coagulation factors, in particular factor II (thrombin), ***the full anti-thrombotic effect of warfarin is not achieved for several days.*** PT may be prolonged soon after administration due to the more rapid reduction of factors with a shorter $t_{1/2}$.

Warfarin: VKORC1 inhibitor
Vit K-dependent factors:
II, VII, IX, X, protein C and protein S

Biochemistry: Biologically inactive prozymogens are converted to active coagulation factors by gamma-carboxylation of N-terminal glutamate residues catalyzed by ***gamma-glutamyl carboxylase with cofactor reduced vitamin K.***

Warfarin binds a VKORC1 subunit → inhibits regeneration of reduced vitamin K from vitamin K epoxide.

Thus, the gamma-carboxylation of prozymogens to the functional zymogens, factors II, VII, XI, X, protein C, and protein S, is inhibited.



Source: Laurence L. Brunton, Björn C. Knollmann: *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:*

VKORC1:
vitamin K epoxide reductase complex subtype 1

Pharmacogenomics: Variable responses

Variability in warfarin response: *CYP2C9* and *VKORC1* genetic variants

---

CYP2C9 reduced function alleles → decreased rate of metabolism → increased risk of bleeding in patients who carry at least one copy of a reduced function allele

---

VKORC1 variations can significantly alter warfarin pharmacodynamics leading to warfarin resistance or warfarin sensitivity1639A or 1173T allele → lower warfarin dose

---

🔑 When determining the initial dose, other patient related factors must also be taken into account; for example: body weight, concomitant medications, and comorbidities.

Warfarin Therapeutic Uses

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Many anticoagulation uses:

- Thromboembolic disorders
  such as, venous thromboembolism and pulmonary embolism
- Prosthetic heart valves
- Atrial fibrillation
- Adjunct to reduce risk of systemic embolism after myocardial infarction
  such as, recurrent MI, stroke
- Acute venous thromboembolism

Bridging with UFH, LMWH, DOAC

Warfarin is not useful for rapid anticoagulation in emergency situations.

- PT / INR measurement is most sensitive to deficiencies in the vitamin K-dependent clotting factors
- Prothrombin time (PT):
  - Measures time for fibrin clot formation
  - Evaluates extrinsic and common coagulation pathways
- International normalized ratio (INR):
  - a calculation for standardizing prothrombin time
  - INR = $[\text{Patient PT}/\text{Normal mean PT}]^{ISI}$
  - Range for patients on warfarin: 2.0 to 3.5 (depending on indication for use)

Warfarin: Adverse Effects

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Effect	Mechanism
Bleeding	Extension of warfarin's mechanism of action • Risk of intracranial hemorrhage increases dramatically with an INR >4 • **especially in older patients**
Thrombosis and skin necrosis	Low level of protein C or protein S • Starting warfarin therapy in patients with acute thrombosis without concomitant heparin or other immediately acting anticoagulant • Inherited deficiency of protein C or protein S
Purple toe syndrome	Cholesterol microembolization • Warfarin therapy may release atheromatous plaque emboli

Warfarin resistance:
Patients requiring >20 mg/day:
- Excessive vitamin K intake
- *VKORC1* variant alleles leading to warfarin resistance

Apparent "resistance"
- Noncompliance / Laboratory error

Warfarin Sensitivity
Patients requiring <1.5 mg/day:
- Reduced function alleles
- *CYP2C9* or *VKORC1* haplotypes
- Lower maintenance dose required
- Increase risk of bleeding

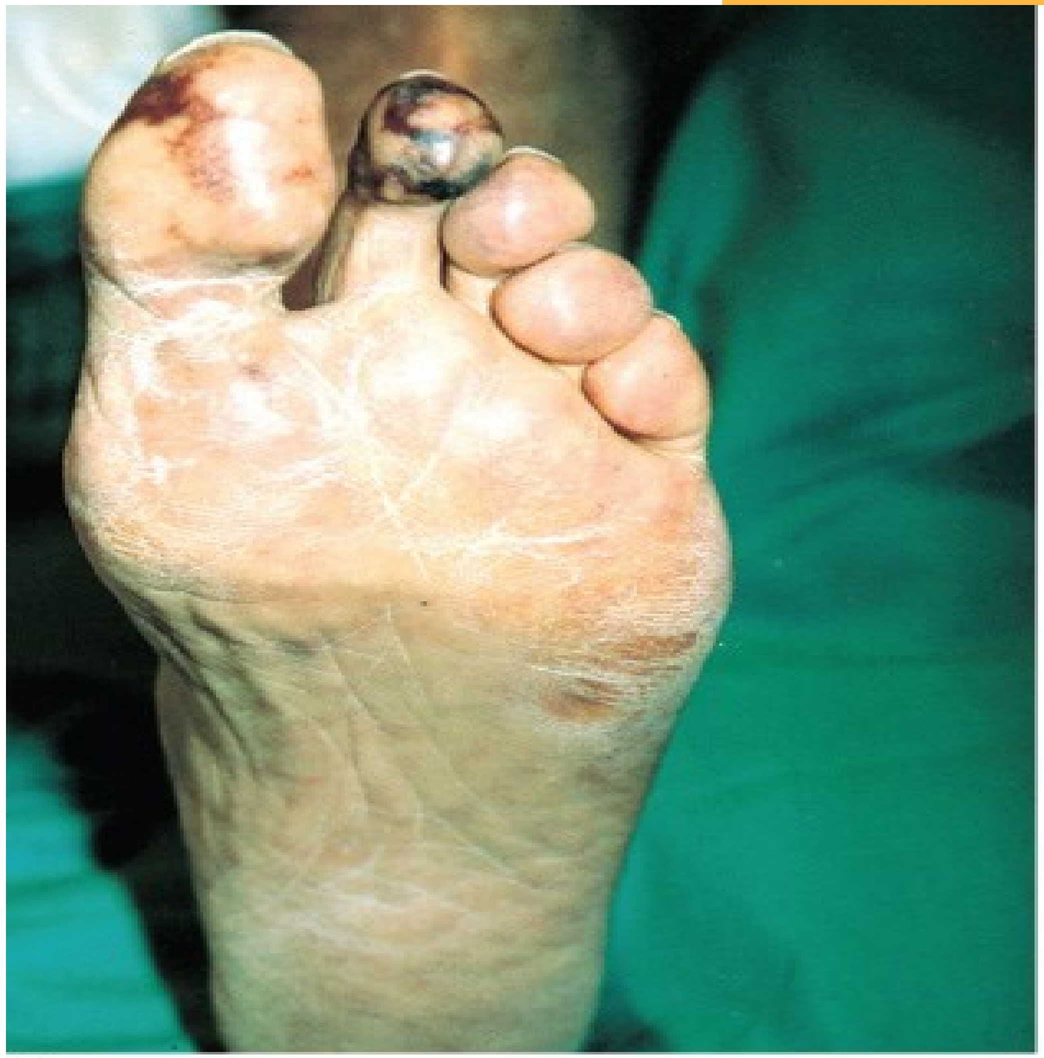
Management of Warfarin Overdose: Hold warfarin

- INR > therapeutic range, no evidence of bleeding: hold drug; restart when INR within range
- INR ≥ 10, no evidence of bleeding: Vitamin K 2.5-5 mg orally; restart when INR within range
- Serious, life-threatening bleeding: IV Vitamin K 10 mg, AND

  4-PCC: Prothrombin complex concentrate

  - Factors II, VII, IX, X, Protein C, Protein S
  - Urgent reversal of acquired coagulation factor deficiency

Skin necrosis in a patient after 4 days of warfarin therapy



A

Purple toe syndrome
(cholesterol emboli)

Left: Access Medicine Fitzpatrick's Dermatology, 8e; Figure 41-6
Right: Access Medicine William's Hematology, 8e; Figure 123-9a, 9b

Warfarin is contraindicated in pregnancy

Fetal warfarin syndrome: nasal hypoplasia and depressed nasal bridge seen in **(A)** a fetal sonographic image and **(B)** in the same newborn

Maternal ingestion during…

<u>First trimester causes:</u>

- Nasal hypoplasia
- Stippling of vertebral and femoral epiphyses

<u>Beyond the first trimester:</u>

- Hemorrhage into fetal structures ⇒ abnormal growth/deformation
- CNS abnormalities
- Spontaneous abortion and fetal death may also occur

Access Medicine William's Obstetrics, 24e, 2014; Chapter 12. Teratology, Teratogens, and Fetotoxic Agents Figure 12-5

Warfarin CONTRAINDICATIONS for Use

- Active bleeding
- Hemorrhagic tendency
- Heparin induced thrombocytopenia
- Severe uncontrolled or malignant hypertension
- Bacterial endocarditis

  These patients are at risk of bleeding complications

- Dissecting aortic aneurysm
- Pericarditis or pericardial effusion
- Lumbar block anesthesia
- Pregnancy
- Hypersensitivity to warfarin or any component of the formulation

Drug Interactions **High potential!** Warfarin is subject to more clinically significant drug interactions than any other drug in use today

- Anticoagulant effect may be:
  - **Reduced**, permitting thrombosis
  - **Increased**, causing hemorrhage
- **The most serious interactions are those that increase the anticoagulant effect**

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Summary

Summary of Anticoagulants

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- Anticoagulants
  - Indirect thrombin inhibitors
    - Heparin
      - UHF
      - LMWH
    - Fondaparinux
  - Direct thrombin inhibitors
    - Argatroban
    - Bivalirudin
    - Desirudin
    - Debigatran
  - Vit K epoxide reductase inhibitors
    - Warfarin
  - Direct Xa inhibitors
    - Rivaroxaban
    - Apixaban
    - Edoxaban
    - Betrixaban

Treatment and prevention of arterial and venous thrombosis

Summary of properties of the oral anticoagulants

FEATURES	WARFARIN	RIVAROXABAN	DABIGATRAN
Bioavailability (%)	100%	80-100%	3-7% (etexilate)
Metabolism	CYP2C9, 1A2, 3A4, 2C19	CYP3A4/5, 2J2 P-glycoprotein substrate	Hepatic glucuronidation P-glycoprotein substrate
t½ (h)	20-60h (highly variable)	5-9h	12-17h
Renal excretion	Yes (mainly metabolites)	Yes, including unchanged Dose adjustment in renal impairment	Yes, largely unchanged Dose adjustment in renal impairment
Antidote effect	Vitamin K	Andexanet-alfa; Not dialyzable	Idarucizumab; ~60% dialyzable
Therapeutic uses	• DVT and PE, prophylaxis and acute treatment • Coronary ischemia/acute MI • Atrial fibrillation, patients with chronic a-fib or prosthetic heart valves	• DVT/PE, prophylaxis/treatment • Non-valvular atrial fibrillation: prophylaxis of stroke and systemic embolism	• DVT / PE prophylaxis/treatment • Non-valvular atrial fibrillation: prophylaxis of stroke and systemic embolism
Drug Interactions	High potential	Drugs that affect platelet function or coagulation CYP3A4, P-glycoprotein related interactions	Drugs that affect platelet function or coagulation P-glycoprotein inhibitors

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References

- Access Medicine Goodman & Gilman The Pharmacological Basis of Therapeutics 14e, 2023; Chapter 36: Blood Coagulation and Anticoagulant, Fibrinolytic, and Antiplatelet Drugs
- Access Medicine William's Obstetrics, 24e, 2014; Chapter 12. Teratology, Teratogens, and Fetotoxic Agents Figure 12-5
- Access Medicine Katzung's Basic and Clinical Pharmacology 16e, 2024; Chapter 34: Drugs Used in Disorders of Coagulation
- Lexidrug monographs

Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R